

Delays and task control

Exercise 1

Implement three tasks where each task controls blinking of one LED. Each task has a dedicated LED and a button. Pressing the button increases LED toggle interval with 100ms increments up to 500ms. When interval is 500ms pressing the button sets the interval to 0 which should set the LED off. When interval is zero pressing the button increases the blinking interval again by 100ms and blinking start again.

Implement the task so that **same function can be used for all three tasks**. Tasks must start with blinking intervals: 100ms, 300ms, and 500ms.

Our board has three buttons that are wired to GPIO pins 7, 8, and 9. The buttons are grounding buttons and need a pull up resistors for reading their state.

There are also three LEDs that are wired to GPIO pins 20, 21, and 22.

Example:

ON – (100ms) – OFF – (100ms) – ON – (100ms)... button press

OFF – (200ms) – ON – (200ms) – OFF – (300ms) – ON – (200ms)... button press

OFF – (300ms) – ON – (300ms) – OFF – (300ms)... button press

ON – (400ms) – OFF – (400ms) ... button press

ON – (500ms) – OFF – (500ms) – ON – (500ms)... button press

OFF – (until button press)... button press

ON – (100ms) – OFF – (100ms) – ON – (100ms)... button press